

WebCrawler

Functional requirements :

user should be to download files based on the metadata ,
user should be able to place multiple downloads

Non-functional requirements:

Data-partitioning,

load balancing,

fault tolerance,

vertical scalability,

horizontal scalability,

low latency,

high throughput,

CAP theorem (Here eventual consistency is fine but here we need 99.99% uptime

i.e Availability>>Consistency)

Core Entities:

user,

file,

filemetadata

API:

POST:/download

{

 userID,

 {fileID}

}

GET:/search/{fileID}

HLD :

